Anirudh Palutla

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Reliable machine learning for protein function: ML-guided directed evolution, sequence-to-function modeling, generalization, and closed-loop design under real experimental constraints.

Education

International Institute of Information Technology, Hyderabad (IIIT-H) 2018 – 2023
B.S. in Computer Science and M.S. in Computational Natural Sciences (dual degree)

- Advisor: Marimuthu Krishnan
- Master's thesis: Investigating Spatial Signatures of Spin-Glass Criticality in the Human Brain

Research experience

Janelia Research Campus (HHMI) Feb 2024 – Present
Scientific Computing Associate · Labs of Srini Turaga and Alison Tebo

- Designed improved GCaMP biosensors through machine learning-guided directed evolution with wet lab validation [preprint in Nov 2026]
- Built and applied a multi-objective optimization loop using sequence-to-function models for biosensor function prediction
- Benchmarked protein foundation models on biosensor fitness prediction, distinguishing sequence-space from fitness-space OOD generalization [ICML 2026 GenBio & FM4LS workshops]
- Modeled pairwise epistasis in iGluSnFR4 with GLMs, finding that interaction strength is inversely correlated with inter-residue distance [Nature Methods, 2026]

IIIT Hyderabad 2020 – 2023
Graduate researcher · Center for Computational Natural Sciences and Bioinformatics

- Built spin-lattice models from resting-state fMRI to compare proximity to criticality in Alzheimer's and cognitively normal brains, finding stretched-exponential temporal correlations consistent with spin-glass rather than Ising-class criticality [Cognitive Neurodynamics, 2024]

Publications

A. Palutla, C. Malin-Mayor, J. N. Koberstein, A. G. Tebo, S. C. Turaga. "Generalization of protein foundation models for engineered fluorescent biosensors." *ICML 2026 Workshop on Foundation Models for the Life Sciences (FM4LS)* and *ICML 2026 Workshop on Generative AI and Biology (GenBio)*, 2026. Poster. OpenReview

A. Aggarwal, A. Negrean, Y. Chen, R. Iyer, D. Reep, A. Liu, **A. Palutla**, et al., J. P. Hasseman, K. Podgorski. "Glutamate indicators with increased sensitivity and tailored deactivation rates." *Nature Methods* 23(2):417–425, 2026. doi:10.1038/s41592-025-02965-z

A. Palutla, S. Seth, S. S. Ashwin, M. Krishnan. "Criticality in Alzheimer's and healthy brains: insights from phase-ordering." *Cognitive Neurodynamics* 18:1789–1797, 2024. doi:10.1007/s11571-023-10033-5

A. Palutla, J. N. Koberstein, A. G. Tebo, S. C. Turaga. "*Working title:* ML-assisted directed evolution of GCaMP calcium indicators." In preparation; preprint to be posted Nov 2026.

Teaching & service

Peer reviewer 2026

- ICML 2026 GenBio Workshop (3 submissions) and ICML 2026 Workshop on Foundation Models for the Life Sciences (3 submissions)

Teaching Assistant, IIIT Hyderabad Spring 2021

- Science II (core course)

Industry experience

Software Development Intern, Walmart Global Tech, Bengaluru May 2022 – Jul 2022

Software Development Intern, AlphaMu Technologies, Hyderabad Aug 2020 – Feb 2021